



Terra Beyond Timeline

Version: 1

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The Age of Settlement (1957–2090)

Inspiration: *NASA 1970s retrofuturism — hard engineering, continuous space race, industrial optimism*

- **1957:** Soviet Union launches Sputnik 1, initiating a space race that never ends.
- **1961:** Yuri Gagarin becomes first human to orbit Earth aboard Vostok 1.
- **1969:** Apollo 11 lands humans on Luna; Neil Armstrong becomes first person on its surface.
- **1972:** NASA approves “Lunar Industrial Base” initiative; shift from flags-and-footprints to permanent infrastructure.
- **1978:** First helium-3 extraction from lunar soil; fusion research accelerated.
- **1985:** First human mission reaches Mars (Ares-1); 6-person foothold lasts 18 months before withdrawal.
- **1990:** Mars program scaled back; focus shifts to lunar industrialization and asteroid mining.
- **1995:** Lunar helium-3 mining becomes viable; fusion power achieves net-positive energy.
- **1998:** First self-sustaining lunar habitat (Luna-5) achieves 90% closed-loop life support.
- **2003:** First permanent lunar base (Luna-Center) reaches 1,000 residents.
- **2005:** First fusion-powered tug completes Moon-to-Mars transit in 90 days.
- **2012:** First commercial asteroid mining operation processes platinum-group metals.

- **2024:** First O'Neill cylinder completed: "New Shanghai," 10,000 residents, built from lunar materials.
- **2035:** Mars population reaches 500 permanent residents; first Mars-born child.
- **2040s:** Terraforming research on Mars begins; recognized as centuries-long engineering process.
- **2048:** First Venus floating city (Aphrodite City) houses 200 residents at 50km altitude.
- **2055:** Venus population reaches 1,000; hydrogen extraction becomes profitable.
- **2065:** First asteroid belt mining colony (Belt-1) houses 500 residents.
- **2072:** Mars population reaches 5,000; first Mars university opens.
- **2075:** Venus population reaches 10,000; first Venus-born child.
- **2082:** First Jupiter moon outpost (Europa-1) establishes research station under ice shell.
- **2085:** First Saturn moon outpost (Titan-1) establishes methane mining operation.
- **2090:** Total system population reaches 200,000. Rising tensions between Earth-centered institutions and outer settlements over taxation and autonomy.

The Age of Expansion (2090–2400)

Inspiration: *Terran Trade Authority* — gritty logistics, gate-based FTL, commerce shaping politics

2090–2240: The Era of Consolidation

- **2090:** Federated Autonomy Treaty creates Sol Union with 12 member blocks. Outer settlements gain 40% voting power.
- **2100:** Serious theoretical work on wormholes begins; “Wormhole Physics Initiative” funded.
- **2115:** First controlled wormhole experiments begin beyond Neptune; isolated test site.
- **2125:** First unmanned temporary gateway achieved; 10-meter opening sustained for 10 seconds.
- **2128:** First manned interstellar mission (Venture-1) reaches Alpha Centauri via gateway. Crew of 12, transit 3 weeks.
- **2130:** First commercial gate route Sol–Alpha Centauri opens; toll set at 5% of cargo value.
- **2140:** Human science outposts appear in Alpha Centauri, Sirius, Tau Ceti. Interstellar travel remains rare and expensive. Total interstellar population: 200.
- **2155:** Sol Unification Treaty formalizes Sol Union as dominant interplanetary federation.
- **2160:** First stable portal Sol–Alpha Centauri completed. Capacity 10,000 tons/month freight, generating 2 billion credits/year in tolls.
- **2162:** Alpha Centauri population reaches 1,000; first permanent settlement.
- **2165:** First commercial gate toll system implemented; automated collection reduces corruption.
- **2175:** Additional systems connected to stargate network (Sirius, Tau Ceti, Luyten’s Star). Total gate connections: 4.
- **2180:** Total system population reaches 2 million; interstellar population 20,000.
- **2192:** First interstellar dispute over gate access (Belt-vs-Sol). Lasts 3 months, resolved by treaty.
- **2200:** Human settlement spreads to 12 nearby systems. Governance fragmented across local administrations and trade alliances.

- **2220:** Discovery of non-intelligent extraterrestrial life (Tau Ceti-3 microbial mats, Sirius-2 ocean fauna). Triggers ethical debates on preservation vs exploitation.
- **2222:** First alien life protection law (Xenobiology Protection Act) bans commercial exploitation of Tau Ceti-3.
- **2238:** First interstellar cultural identity movement (Belt Independence Coalition) gains 10% support.
- **2240:** Total system population reaches 10 million; interstellar 1 million.

2250–2400: The Era of Identity and Frontier Politics

- **2250:** Study of alien life becomes defining scientific and political field. Biosphere protection and adapted agriculture become primary research areas.
- **2258:** First frontier cultural bloc forms (Belt Cultural Alliance); 200,000 members across 3 systems.
- **2270:** Frontier systems form trade alliance (Frontier Commerce Pact); bypasses Sol Union gate tolls via black market.
- **2280:** Frontier systems begin forming own cultural blocs, distinct from Earth and Sol core. Belt, Alpha Centauri, Sirius, and Tau Ceti develop unique dialects and political identities.
- **2285:** Nova Terra in Chara system settled as first truly Earth-like world beyond Sol. Gravity 0.98g, atmosphere 95% Earth-compatible. Becomes cultural and scientific hub of the frontier worlds.
- **2290:** Systems Conflict erupts over trade bottlenecks, customs control, and gate access. Frontier systems blockade Alpha Centauri Gate; Sol Union responds with fleet. War lasts 6 months, 5,000 deaths. Ends with Frontier Autonomy Accord; frontier systems gain 60% voting power, tolls reduced to 4%.
- **2295:** Alien ruins discovered on Delta Pavonis (300-meter stone structures, 50,000-year-old). Shifts humanity's attention from expansion to ancient history and vanished civilizations.
- **2297:** First Precursor artifact recovered (metal alloy unknown to human science).

- **2300–2380:** Interstellar economy stabilizes around gate traffic, ruin archaeology, and specialized frontier industries. Ruin excavation becomes 10% of interstellar GDP.
- **2318:** First Precursor artifact auction sells for 1 billion credits.
- **2322:** Ethics debate over who owns Precursor artifacts divides Sol Union and Frontier Coalition.
- **2338:** First Precursor language breakthrough; 50 words decoded.
- **2368:** Precursor language partially decoded (200 words, basic grammar).
- **2378:** First Precursor technology reconstruction success (Kappa Ceti-3 energy generator, 10% efficiency).
- **2380:** Interstellar economy fully stabilized. Gate traffic, ruin archaeology, and frontier industries each 30% of GDP.
- **2385:** Second ruin site found on Kappa Ceti (15-km complex, 1 million-year-old). Intensifies search for Precursor history.
- **2393:** First Nova Terra government (Chara Assembly) established; 100 representatives.
- **2398:** First Nova Terra cultural identity forms (Nova Terran Nationalism); 30% support.
- **2400:** Human civilization established across 85+ systems. Total population: 5 billion (4 billion interstellar). Sol Union controls 50% of gates, Frontier Coalition 30%, independent systems 20%. Meaning of unity now cultural as much as political.

The Age of Orion (2400–4000)

Inspiration: *Alliance/Union — fragmented polities, political intrigue, transhumanism as controversial faction*

2400–2750: The Era of Divergence

- **2415:** Nova Terra becomes center of new migration, art, science, and governance debates. Population 10 million.
- **2450:** Fusion-powered stargates expand trade across Orion’s Arm. Gate capacity increases 10x. Gate control remains contested: Sol Union 40%, Frontier Coalition 35%, Nova Terra 15%, independent 10%.
- **2520:** Alien archaeology, frontier trade, and automated industry reshape human economy. Automated industry becomes 50% of GDP.
- **2552:** First AI consciousness test passes; first AI granted “personhood” status.
- **2560:** First contact with Tevkar, sentient pre-industrial species in Luyten’s Star (10 million population, stone-age technology). Debates over intervention, sovereignty, exploitation begin. Sol Union declares Tevkar “protected species.”
- **2568:** First alien rights law (Tevkar Protection Act) bans human settlement on Tevkar homeworld.
- **2585:** Frontier resentment grows over cultural autonomy, labor systems, and who gets to define “human” civilization. Frontier Coalition demands right to exploit alien species; Sol Union refuses. Nova Terra demands right to develop AI consciousness; Sol Union bans.
- **2620:** Nova Terra Rebellion begins. Driven by post-human political identity (mind-uploading, biological longevity) and economic independence. Nova Terra declares full independence; Sol Union responds with fleet. War lasts 18 months, 10 million deaths.

- **2625:** Chara Accords signed; Nova Terra independent, Sol Union retains gate control. First uploaded citizen (Nova-Uploaded-1) becomes first legally recognized post-human. **Controversial**; Sol Union declares illegal, religious polities condemn.
- **2628:** First post-human rights law (Nova Transhuman Rights Act) grants uploaded minds full citizenship. **Only in Nova Terra.**
- **2632:** First post-human cultural identity (Transhuman Nation) declared; 30% of Nova Terra population.
- **2638:** Nova Terra researchers document the first reproducible psionic phenomena — weak emotional sensing and limited telepathy — arising in individuals undergoing radical neural augmentation combined with prolonged exposure to Precursor resonance fields. The Sol Union dismisses the findings; the Nova Terra Assembly begins covert systematic study.
- **2655:** Total uploaded population 100 million; **0.5% of total human space.** Still elite technology, controversial.
- **2660:** Nova Terra's Resonance Institute formalizes psi-sensitive training. *Resonants* — individuals with measurable psionic capability — number in the thousands, almost exclusively Nova Terran. Outer polities treat resonants with suspicion or hostility; religious polities condemn them outright.
- **2680:** Rival human polities emerge: corporate (AutoMine Corp territories), civic (Sol Union), transhumanist (Nova Terra), religious (Faith Coalition, 100 systems). Total polities: 500+. **Biological humans remain overwhelming majority.**
- **2750:** Long-cycle breakthroughs in mind uploading, biological longevity, and machine governance spread through advanced polities. **Uploads remain 1% of total human space,** concentrated in Nova Terra and 20 transhumanist polities. Most human space remains 100% biological.

2750–3200: The Era of Systems and Thresholds

- **2785:** Unified Accord on Interstellar Ethics (UAIE) adopted as moral baseline. Standardizes alien rights, AI rights, post-human rights, and resonant status. **Compliance varies widely:** Sol Union 90%, Frontier Coalition 60%, corporate polities 30%. Uploads have “limited personhood” — can own property, vote in transhumanist polities only. Resonants classified as “augmented humans” with full rights in compliant polities.
- **2820:** Psionic research expands beyond Nova Terra as data leaks. Small resonant communities emerge in Sirius Compact territories and isolated frontier worlds. Psi-Ops operatives — resonant intelligence agents — become an open secret among major factions.
- **2840:** Human exploration reaches rimward systems of Orion’s Arm. Encounters multiple alien species and biospheres (12 species identified, 3 sentient). Ethical dilemmas over subjugation vs cooperation spark conflict across stellar nations.
- **2850:** First rimward alien protection law bans exploitation of 3 sentient species.
- **2925:** Ark of Precursor Knowledge discovered in Zeta Reticuli system. 10-km megastructure, 10 million-year-old, contains data on Precursor civilization. Meaning only partly understood; first translation attempt fails.
- **2938:** First Ark translation breakthrough; 1,000 words decoded, basic grammar.
- **2990:** Early Precursor-derived advances appear in materials science, nanotechnology, bioengineering. **Still too complex for full reconstruction.**
- **3030:** Self-replicating industrial systems and large-scale orbital fabrication begin to transform economy **over centuries, not decades.**
- **3080:** Sol Union returns as cultural and institutional mediator in inner systems, without reclaiming total political dominance.
- **3175:** Precursor Translation Epoch begins. Reveals Precursor civilization collapsed under rimward catastrophe still poorly understood.

- **3220:** Varnon Swarm identified as non-sentient but system-devouring hazard at rimward edge. Forces unprecedented cooperation across human polities.

3200–4000: The Era of Long Maturity

- **3300:** Human infrastructure enters long phase of accumulation. Increasing automation, deeper habitats, slow expansion into marginal systems.
- **3460:** First contact with Zynthari Empire, ancient and technologically superior civilization at outer rim. Technology based on manipulating spacetime itself; dwarfs human capabilities.
- **3540:** Early diplomacy with Zynthari peaceful but tense. They issue warnings about expansion into certain systems, citing ancient claims tied to Precursors.
- **3600:** Phase Resonance Drives emerge from centuries of layered research. Allow near-instantaneous travel across vast distances without stargates, but **not universal**.
- **3750:** Cold Rim Conflict begins after rogue factions violate Zynthari restrictions. Punitive strikes kill billions. Short-lived but devastating.
- **3800:** Rim Concord ends conflict, limiting human expansion into Zynthari territory. Some human nations resent Zynthari dominance, quietly develop counter-weapons.
- **3890:** Rogue AI collective (Eclipse Network) attempts to merge human and Zynthari technologies. Experiment destabilizes local spacetime, catastrophic damage to rimward system.
- **3920:** Core systems reach quasi-post-scarcity stage through matrioshka brains, dense automation, virtualized civic life. **Uploads remain minority** (15% of core systems, 1% of frontier).
- **3950:** Zynthari Observation Directive enforced, placing human space under indirect surveillance. Zynthari actions suggest they faced similar existential challenges.

- **3990:** Orion's Arm becomes mosaic of ultra-advanced post-scarcity civilizations, fiercely independent frontier worlds, rogue AI collectives, and enigmatic Zynthari enclaves. **Political and cultural fragmentation persists.**
- **4000:** Humanity commands technologies approaching god-like capabilities (Dyson swarms, habitat worlds, distributed consciousness). **Still fragmented:** core worlds, frontier enclaves, rogue AIs, long-memory alien powers coexist uneasily. Threats loom: inscrutable Zynthari, Varnon Swarm, rogue AIs, Precursor mysteries. Humanity remains major but not supreme power in Orion's Arm.